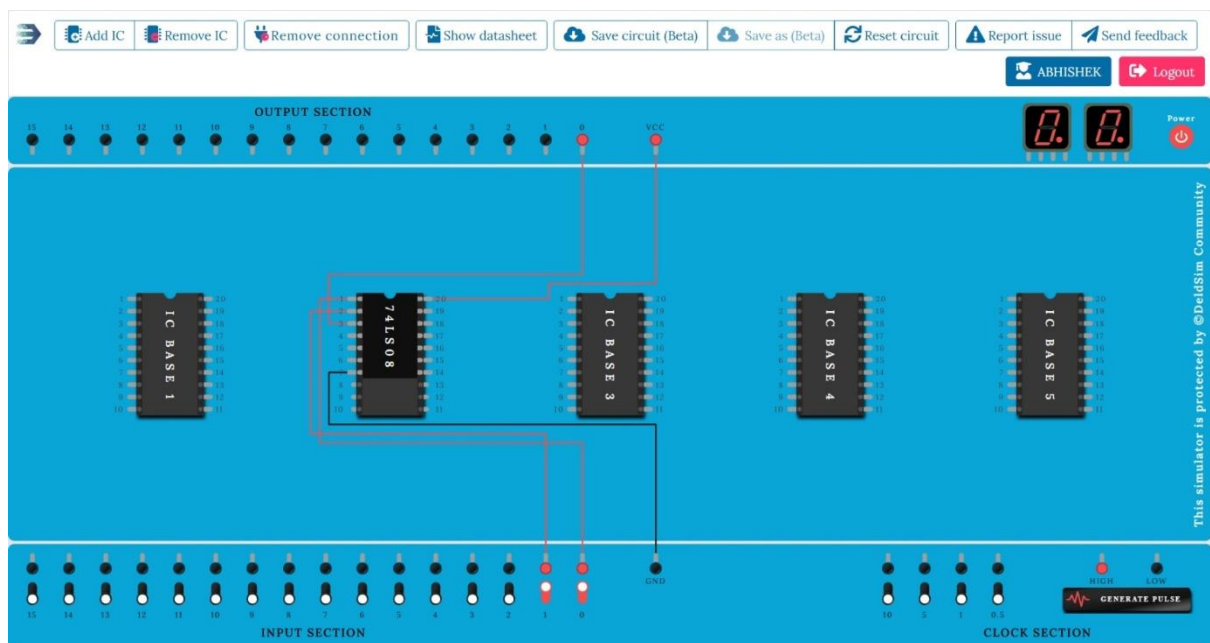


Assignment:- 3

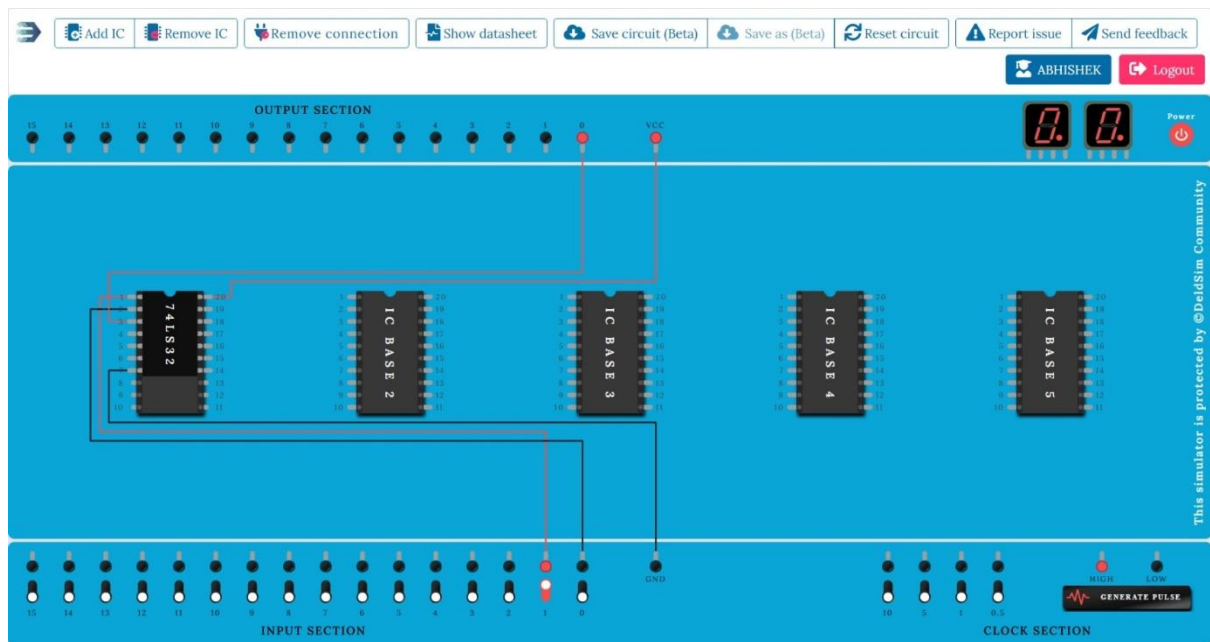
Verification of Logic Gates and Boolean Algebra

The following are the logic gates and their outputs-

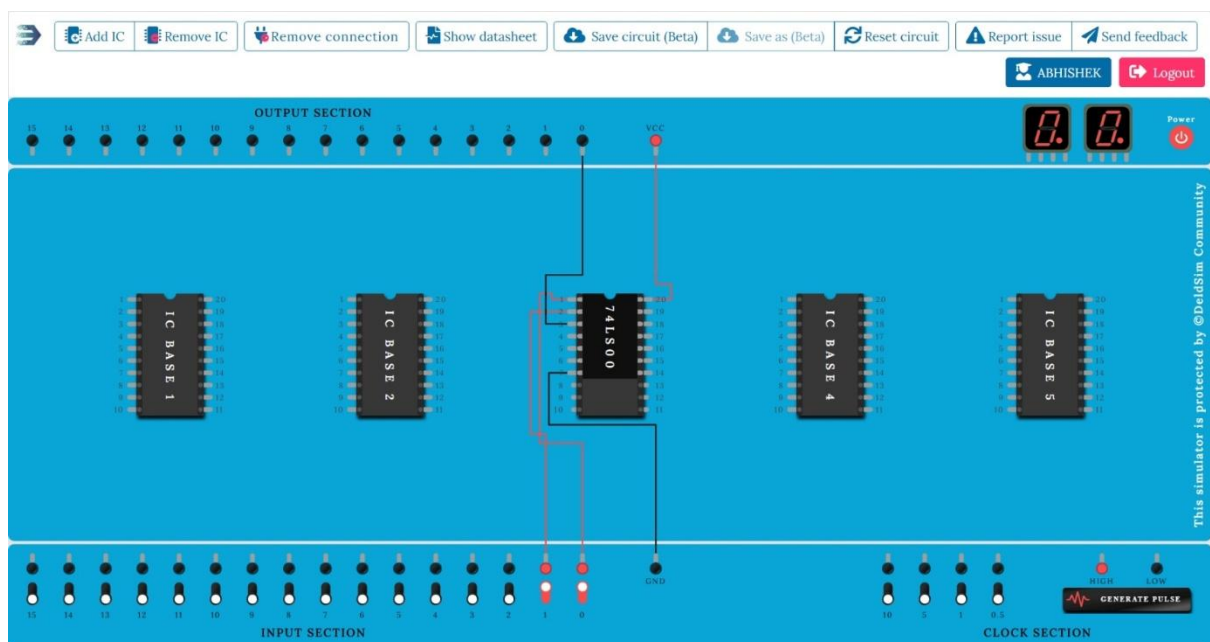
AND gate:-



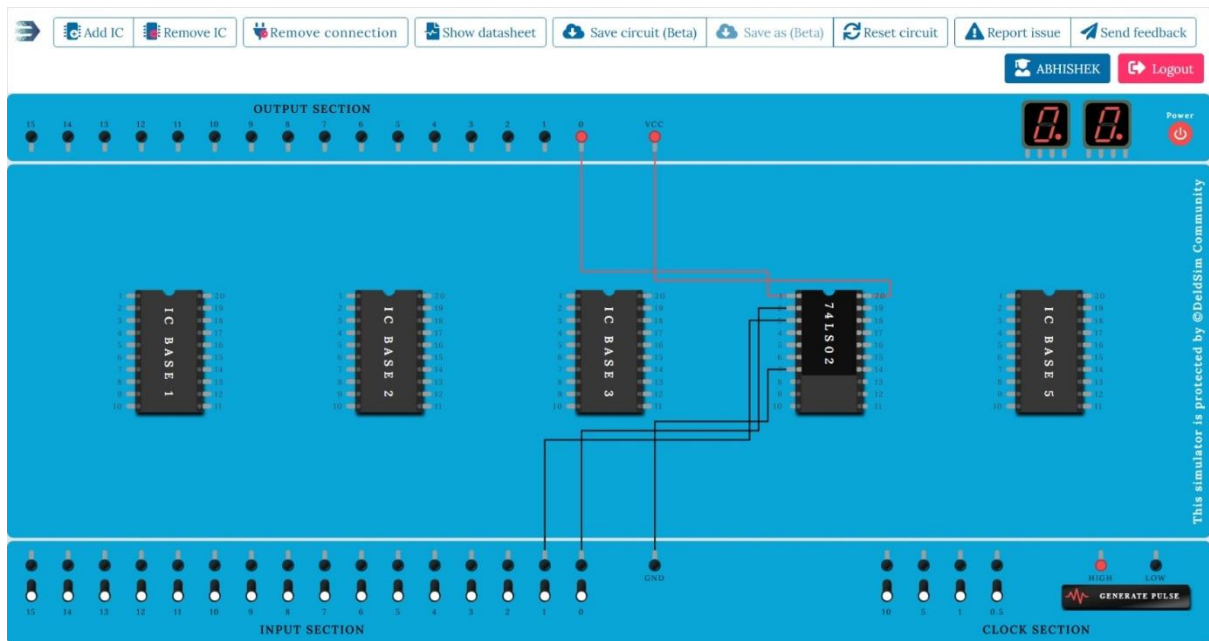
OR gate:-



NAND gate:-



NOR gate:-



Laws for Boolean algebra:-

(a) Associative Law:-

- States that the grouping of variables does not affect the result.
- For AND ($\cdot$):
$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$
- For OR ($+$):
$$(A + B) + C = A + (B + C)$$

(b) Distributive Law:-

- States that multiplication can be distributed over addition and vice versa.
- $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

- $A+(B \cdot C)=(A+B) \cdot (A+C)$
 $A+(B \cdot C)=(A+B) \cdot (A+C)$
 $(B \cdot C)=(A+B) \cdot (A+C)$

(c) Absorption Law:-

- Simplifies expressions by "absorbing" terms.
- $A+(A \cdot B)=A$
 $A+(A \cdot B)=A$
- $A \cdot (A+B)=A$
 $A \cdot (A+B)=A$

De Morgan's Theorem:-

Used to simplify Boolean expressions involving NOT operations.

- $(A \cdot B)'=A'+B'$
 $(A \cdot B)'=A'+B'$
 (Complement of AND = OR of complements)
- $(A+B)'=A' \cdot B'$
 $(A+B)'=A' \cdot B'$
 (Complement of OR = AND of complements)

Principle of Duality:-

- Every Boolean equation remains valid if we interchange + and $\cdot$ and replace 0 with 1 and 1 with 0.
- Example:
 - Original: $A+0=A$
 $A+0=A$
 - Dual: $A \cdot 1=A$
 $A \cdot 1=A$

Minterm and SOP (Sum of Products):-

Minterm:

- A product (AND) term where each variable appears once (either complemented or uncomplemented).
- Example for 2 variables:
 - $M_0 = A'B'$
 - $M_1 = A'B$
 - $M_2 = AB'$
 - $M_3 = AB$

Sum of Products (SOP):

- Expression is the sum (OR) of selected minterms.

A	B	Minterm (Product)	Decimal Index
0	0	$A'B'$	m0
0	1	$A'B$	m1
1	0	AB'	m2
1	1	AB	m3

- Example SOP:

$$F(A,B)=m_1+m_3=A'B+AB$$
$$F(A,B)=m_1+m_3 = A'B + AB$$
$$F(A,B)=m_1+m_3=A'B+AB$$

Maxterm and POS (Product of Sums):-

Maxterm:

- A sum (OR) term where each variable appears once (either complemented or uncomplemented).
- Example for 2 variables:
 - $M_0 = A + B$
 - $M_1 = A + B'$
 - $M_2 = A' + B$
 - $M_3 = A' + B'$

Product of Sums (POS):

- Expression is the product (AND) of selected maxterms.

A	B	Maxterm (Sum)	Decimal Index
0	0	$A + B$	M_0
0	1	$A + B'$	M_1
1	0	$A' + B$	M_2
1	1	$A' + B'$	M_3

- Example POS:

$$F(A,B)=M_0 \cdot M_2 = (A+B)(A'+B) \quad F(A,B) = M_0 \cdot M_2 = (A+B)(A'+B)$$